

Relating consumer problem to how univariate optimization problem with single inequality constraint looks in book's appendix:

$$\max_x f(x) \text{ subject to } h(x) \leq c$$

KT conditions: $f'(x) = \lambda h'(x)$ and $\lambda(h(x) - c) = 0$. Here, $h(x) = -x$ and $c = 0$:

$$\max_{x_i} \phi_i(x_i) - p^* x_i \text{ subject to } x_i \geq 0$$

KT conditions: (i) $\phi'_i(x_i) - p^* = -\lambda$ and $\lambda x_i = 0$. So $\phi'_i(x_i) - p^* + \lambda = 0$, where $\lambda = 0$ if $x_i > 0$ and $\lambda > 0$ if $x_i = 0$ (so $\phi'_i(x_i) \leq p^*$).